

Linear Regression Predicts a Continuous Target

Linear regression is a supervised algorithm for predicting a continuous numerical target with a weighted sum of features.



CORE CONCEPT

The Linear Regression Equation Has Four Parts

For n features: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$ Each term $\beta_i x_i$ is one feature's contribution. The model adds all contributions and the intercept.

$\hat{y}$ (y-hat): predicted target

β_0 : intercept, the prediction when all features equal zero

x_i : value of feature i

β_i : coefficient for feature i

n : number of features

RUN IT AND INSPECT THE RESULT

A Coefficient Describes a Conditional Change

```
def predict_price(area):  
    return 50000 + 120 * area  
  
print(predict_price(100))
```

OUTPUT

62000

For area 100: $50,000 + 120(100) = 62,000$.

REASONING SEQUENCE

Multiple Features Add Separate Contributions

Frame

With several features, linear regression can be written:

1

Transform

$x \cdot \beta$ is the dot product:

2

Validate

Feature order must match coefficient order.

3

Explain

Changing units changes coefficient size, which is why raw coefficient magnitudes are not automatically comparable.

4

Model: $\hat{y} = 20 + 3(\text{distance}) - 2(\text{items})$ For distance 5 and items 4: $\hat{y} = 20 + 3(5) - 2(4) = 20 + 15 - 8 = 27$ minutes

CORE CONCEPT

Residuals Show Individual Prediction Errors

A residual is actual minus predicted: $e_i = y_i - \hat{y}_i$ Residual plots help reveal patterns that a single score hides. A useful linear fit usually leaves residuals scattered around zero without a clear curve or funnel shape.

positive residual: the model predicted too low;

negative residual: the model predicted too high;

zero residual: prediction matched the observed target.

RUN IT AND INSPECT THE RESULT

MSE and RMSE Summarize Residuals

```
from sklearn.metrics import mean_squared_error

mse = mean_squared_error(actual, predicted)
rmse = mse ** 0.5
print(mse, round(rmse, 2))
```

OUTPUT

7.0 2.65

Squared: [4, 16, 1] Sum: 21 MSE = $21 / 3 = 7$ RMSE = $\sqrt{7} \approx 2.65$

CORE CONCEPT

Ordinary Least Squares Chooses the Best Coefficients

Ordinary least squares (OLS) chooses coefficients that minimize the sum of squared residuals: $\min_{\beta} \sum (y_i - \hat{y}_i)^2$. Each candidate line produces residuals. Squaring and summing gives one cost. OLS selects the coefficients with the smallest cost among the candidates considered by the solver.

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With one feature, the result is the fitted line.

With many features, it is a fitted plane or higher-dimensional hyperplane.

REASONING SEQUENCE

Gradient Descent Can Minimize the Same Loss

Frame

start with coefficient values;

1**Transform**

predict training targets;

2**Validate**

calculate MSE;

3**Explain**

calculate gradients;

4

Current coefficient $\beta = 2$, gradient -6 , learning rate 0.1 : $\beta_{\text{new}} = 2 - 0.1(-6) = 2 + 0.6 = 2.6$ The negative gradient means increasing the coefficient locally reduces loss.

RUN IT AND INSPECT THE RESULT

Assumptions Tell Us When the Model Is Credible

```
residuals = y_test - model.predict(X_test)
plt.scatter(model.predict(X_test), residuals)
plt.axhline(0, color="black")
```

OUTPUT

A residual plot centred on a horizontal zero line.

If residual spread grows from about ± 2 to ± 20 as predictions increase, variance is not constant. One RMSE value hides that the model is much less reliable for large predictions.

QUALITY GATE

Evaluate Against a Mean Baseline on Unseen Data



SSres: squared residual sum for the model



SStot: squared deviation sum from the target mean

QUALITY GATE

Ridge and Lasso Control Coefficient Size



SSE: sum of squared residuals



α : penalty strength



Ridge usually shrinks coefficients toward zero.



Lasso can set some coefficients exactly to zero.

GUIDED LAB

Guided Lab: Explain Every Part of a Regression

01

**observation, features,
and continuous target**

02

**baseline test
RMSE**

03

**fitted equation
with units**

04

**one prediction
calculated by hand**